

LAB EXPERIMENT RECORD

SET 4: Product Inventory Management

Department	AIDS
Course Code / Course Name	DSA0602 / Data Handling and Visualization
Experiment	Data Visualization using R
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Day	Thursday

Aim

To analyze product inventory data using different data visualization techniques such as bar chart, stacked bar chart, scatter plot, and an interactive dashboard.

Dataset: Product Inventory

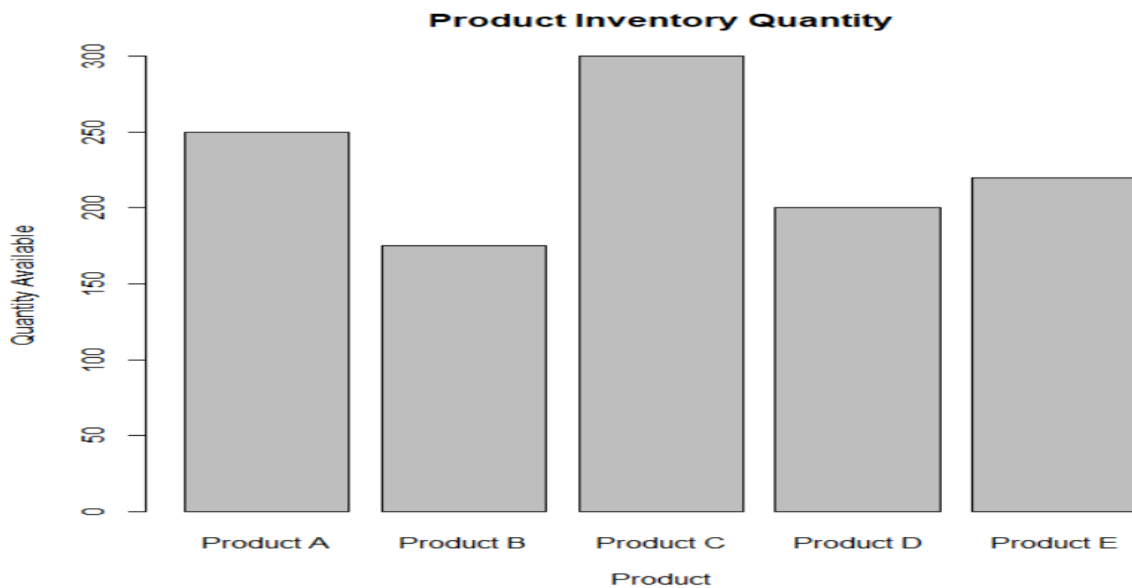
Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Program 1: Product Inventory Bar Chart

Code

```
product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")  
quantity <- c(250, 175, 300, 200, 220)  
inventory <- data.frame(product, quantity)  
barplot(inventory$quantity,  
        names.arg = inventory$product,  
        main = "Product Inventory Quantity",  
        xlab = "Product",  
        ylab = "Quantity Available")  
text(x = seq_along(inventory$quantity),  
     y = inventory$quantity,  
     labels = inventory$quantity,  
     pos = 3)
```

Output:



Observation

Product C has the highest quantity available with **300 units**, while Product B has the lowest quantity with **175 units**.

Program 2: Stacked Bar Chart by Product Category

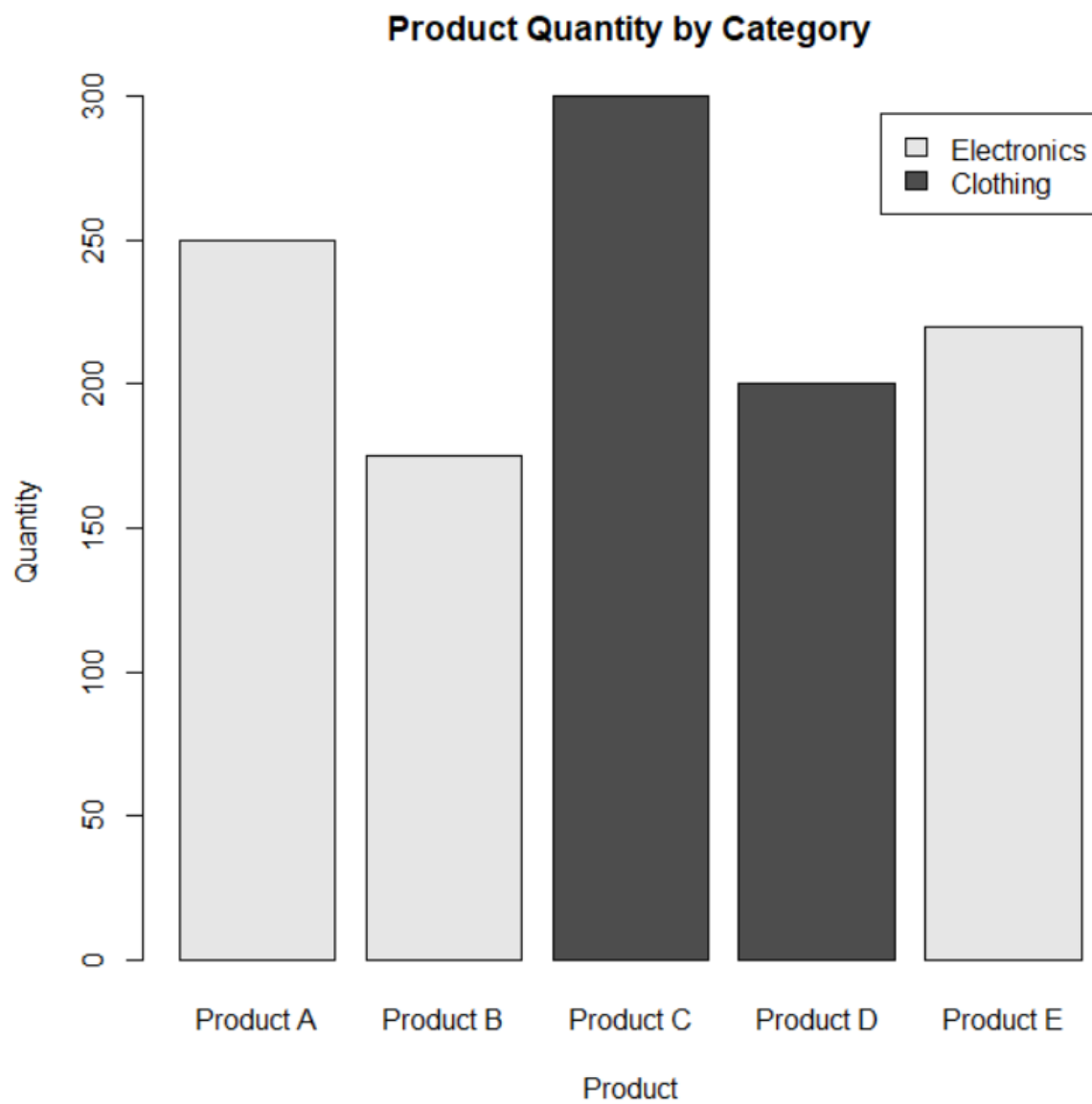
Additional Sample Data

Product	Category	Quantity
Product A	Electronics	250
Product B	Electronics	175
Product C	Clothing	300
Product D	Clothing	200
Product E	Electronics	220

Code

```
product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")  
category <- c("Electronics", "Electronics", "Clothing",  
             "Clothing", "Electronics")  
quantity <- c(250, 175, 300, 200, 220)  
inventory <- data.frame(product, category, quantity)  
data_matrix <- xtabs(quantity ~ product + category,  
                    data = inventory)  
barplot(t(data_matrix),  
        beside = FALSE,  
        main = "Product Quantity by Category",  
        xlab = "Product",  
        ylab = "Quantity",  
        legend.text = TRUE)
```

Output:



Observation

The stacked bar chart shows the quantity of each product according to its category. It makes comparison between **products and categories** easier.

Program 3: Scatter Plot – Product Price vs Quantity

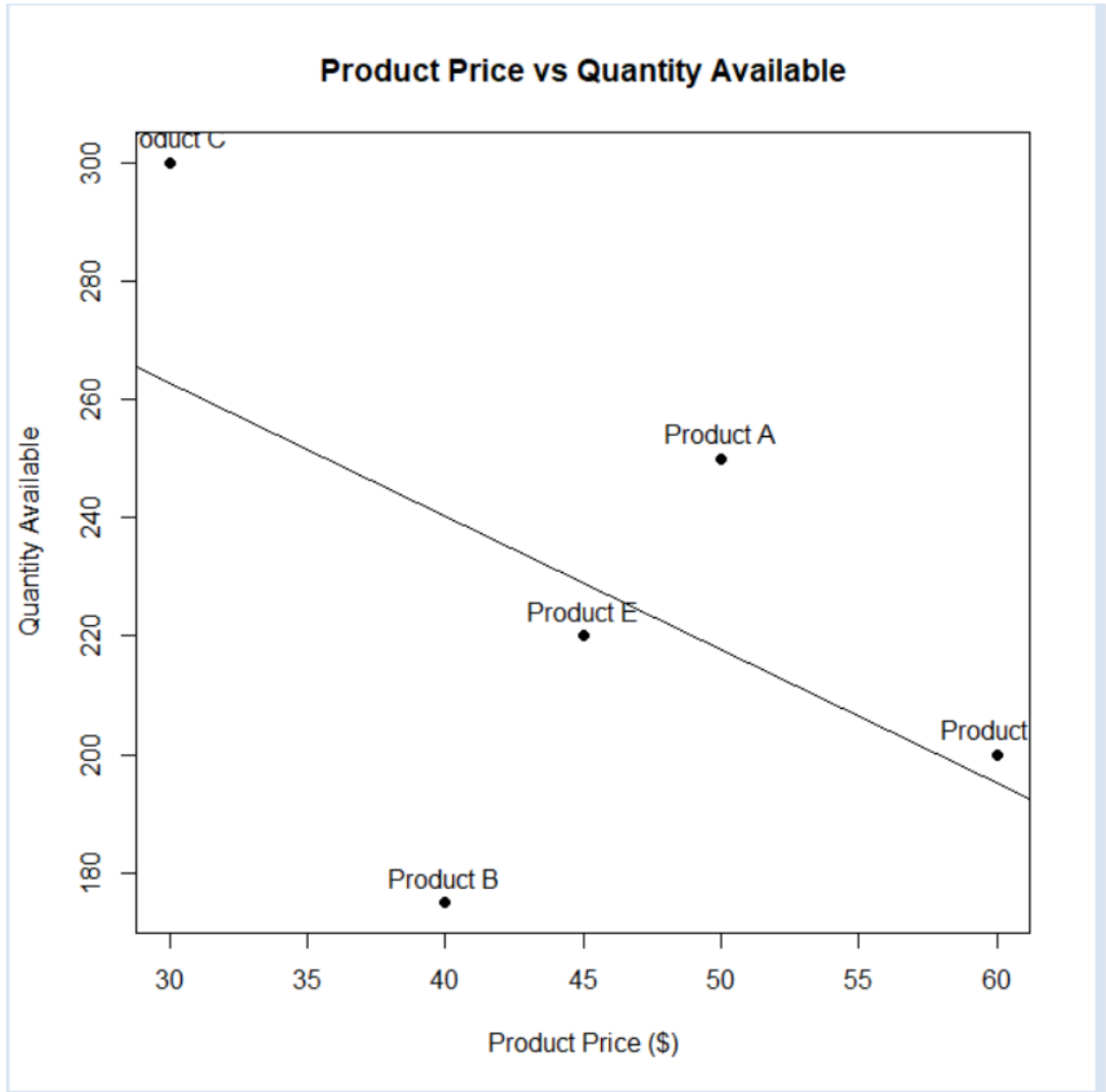
Additional Sample Data

Product	Price (\$)	Quantity
Product A	50	250
Product B	40	175
Product C	30	300
Product D	60	200
Product E	45	220

Code

```
product <- c("Product A", "Product B", "Product C",  
            "Product D", "Product E")  
price <- c(50, 40, 30, 60, 45)  
quantity <- c(250, 175, 300, 200, 220)  
inventory <- data.frame(product, price, quantity)  
plot(inventory$price,  
     inventory$quantity,  
     main = "Product Price vs Quantity Available",  
     xlab = "Product Price ($)",  
     ylab = "Quantity Available",  
     pch = 19)  
text(inventory$price,  
     inventory$quantity,  
     labels = inventory$product,  
     pos = 3)  
  
abline(lm(quantity ~ price,  
          data = inventory))
```

Output:



Observation

The scatter plot shows a general negative relationship between product price and quantity available in the sample data. Higher-priced products tend to have comparatively lower quantities.

Program 4: Tableau Dashboard

Code

```
library(shiny)

product <- c("Product A",
             "Product B",
             "Product C")

january <- c(2000, 1500, 1200)

february <- c(2200, 1800, 1400)

march <- c(2400, 1600, 1100)

sales_data <- data.frame(

  Product = product,

  January = january,

  February = february,

  March = march)

ui <- fluidPage(

  titlePanel("Product Sales Analysis Dashboard"),

  mainPanel(

    h3("Grouped Bar Chart"),
```

```
    plotOutput("barPlot"),  
  
    h3("Sales Trend"),  
  
    plotOutput("areaPlot"),  
  
    h3("Monthly Sales Table"),  
  
    tableOutput("salesTable")  
  
  )  
  
)
```

```
server <- function(input, output) {  
  
  output$barPlot <- renderPlot({  
  
    sales <- rbind(  
  
      sales_data$January,  
  
      sales_data$February,  
  
      sales_data$March  
  
    )  
  
    barplot(  
  
      sales,  
  
      beside = TRUE,
```

```
names.arg = sales_data$Product,  
  
main = "First Quarter Product Sales",  
  
xlab = "Product",  
  
ylab = "Sales",  
  
legend.text =  
  
  c("January",  
  
    "February",  
  
    "March")  
  
)  
  
})  
  
output$areaPlot <- renderPlot({  
  
  sales <- rbind(  
  
    sales_data$January,  
  
    sales_data$February,  
  
    sales_data$March  
  
  )  
  
  
  
  matplot(  
  
    1:3,
```

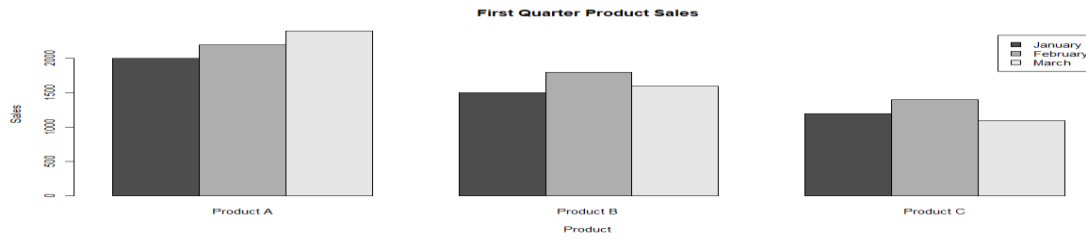
```
sales,  
  
type = "l",  
  
lty = 1,  
  
lwd = 2,  
  
xaxt = "n",  
  
main = "Product Sales Trend",  
  
xlab = "Month",  
  
ylab = "Sales"  
  
)  
  
axis(  
  
1,  
  
at = 1:3,  
  
labels =  
  
c("January",  
  
  "February",  
  
  "March")  
  
)  
  
legend(  
  
  "topleft",
```

```
    legend = sales_data$Product,  
  
    lty = 1  
  
  )  
  
  })  
  
  
  
  output$salesTable <- renderTable({  
  
    sales_data  
  
  })  
  
}  
  
shinyApp(ui = ui, server = server)
```

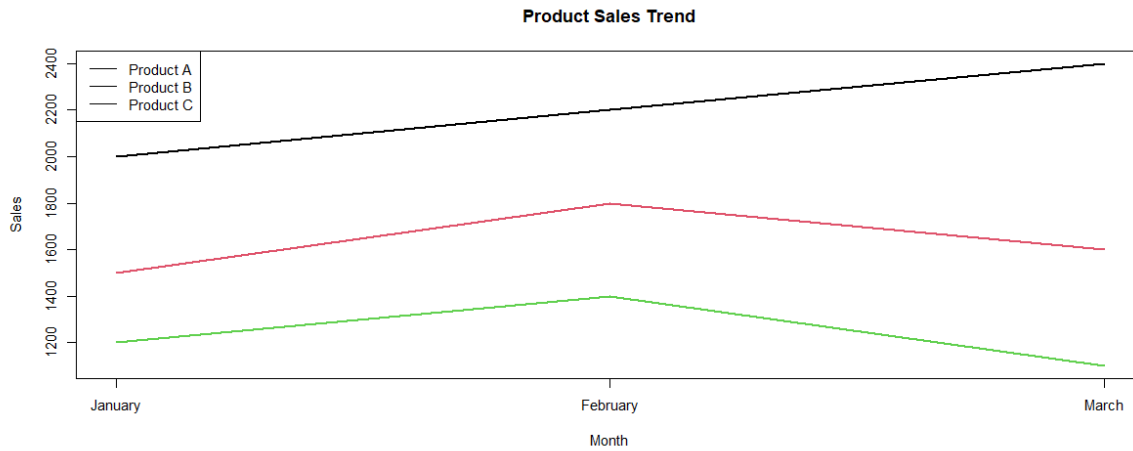
Output:

Product Sales Analysis Dashboard

Grouped Bar Chart



Sales Trend



Monthly Sales Table

Product	January	February	March
Product A	2000.00	2200.00	2400.00
Product B	1500.00	1800.00	1600.00
Product C	1200.00	1400.00	1100.00

Observation

The Tableau dashboard combines the inventory bar chart and category-wise stacked bar chart, allowing users to interactively explore product quantities.

Procedure

1. Enter the product inventory dataset.
2. Create a data frame in R.
3. Create a bar chart to visualize product quantities.

4. Add category information and create a stacked bar chart.
5. Add product price information and create a scatter plot.
6. Analyze the relationship between price and quantity.
7. Import the inventory data into Tableau.
8. Create the bar chart in Tableau.
9. Create the stacked bar chart in Tableau.
10. Combine both charts into an interactive dashboard.
11. Add filters and labels to improve dashboard interaction.

Result

The product inventory data was successfully analyzed using bar chart, stacked bar chart, and scatter plot techniques. An interactive Tableau dashboard was also developed by combining the bar chart and stacked bar chart for product inventory exploration.

Key R Functions Used

R Function	Purpose
c()	Creates vectors
data.frame()	Creates data frames
barplot()	Creates bar and stacked bar charts
text()	Adds values/product labels to charts
xtabs()	Creates a contingency/table matrix
plot()	Creates scatter plot
lm()	Creates linear regression model

abline()	Adds regression line
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